Minor Project (21CSA697A)

<u>Proposal(Individual mode)</u>

Title : Pneumonia detection from chest x rays using deep learning.
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Abstract:

This project focuses on detecting pneumonia from chest x rays using deep learning techniques. A convolutional neural network model will be developed using pretrained architectures such as VGG or ResNet and fine tuned on a chest x-ray dataset. Image preprocessing and data augmentation will be applied to improve accuracy. Additionally explainable AI methods like Grad-CAM will be used to highlight important regions in the x-rays influencing the model's predictions. The system aims to provide a fast, accurate and interpretable solution to assist in medical diagnosis.

Assumptions/Declarations:

The chest x-ray dataset used is accurate and properly labeled. Pre-trained deep learning models can improve the performance of pneumonia detection. The accuracy of the system depends on data quality and preprocessing techniques. The proposed system is designed to assist healthcare professionals, not replace them. Required tools and computational resources are available for implementation.

Main objective/ Deliverable:

To design and implement a deep learning-based system for pneumonia detection from chest x-ray images. To use pre-trained CNN models for improving classification accuracy. To incorporate explainable AI techniques for better interpretation of results. To develop a simple interface for demonstrating the model's predictions.

Timeline and Milestones:

	Milestones	Timeline
1.	Problem understanding and dataset collection Data processing and augmentation Model selection and training Model evaluation and tuning Implementation Testing and final report submission	Week 1-2 Week 3-4 Week 5-6 Week 7-8 Week 9-10 Week 11 -12

Tools to be used

Software/Hardware tools	**Specifications**
Python Tensorflow/keras/Pytorch Opencv, numpy,pandas matplotlib/seaborn	Programming language Model building and training Data processing Visualisation

Learning involved

Topic	**Description**
Deep learning Transfer learning Image processing Model evaluation Explainable AI	Understanding image classification models. Using pre-trained models Handling and preprocessing x-ray images Accuracy,loss, performance metrics Interpreting model predictions.

Date	27 april 2026
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